IGCSE 0607 Practice Questions — E4.2

Gradient of linear graphs

Total Questions: 18 — **Total Marks:** 107 Marks

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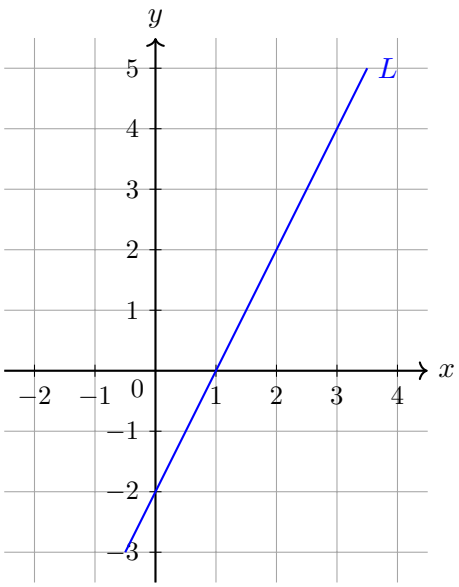
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1 Points A and B have coordinates $(2, -3)$ and $(6, 5)$ respectively.

Work out the gradient of the line AB.

..... **[2]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

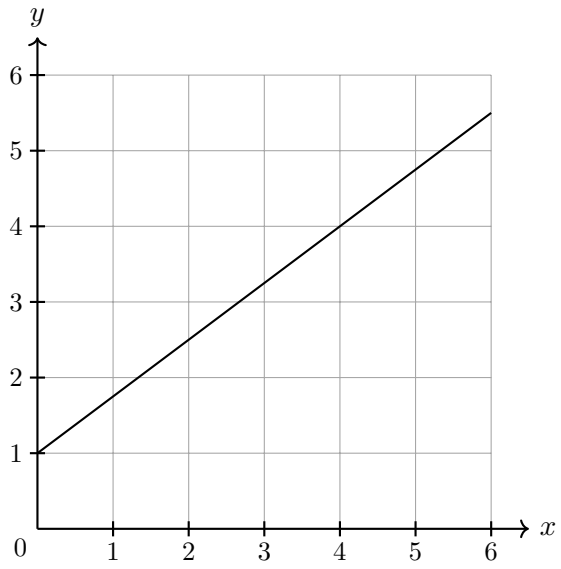
(..... , ...) **[1]**

3 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

4 The points $A(-4, 7)$ and $B(2, -5)$ lie on a straight line.

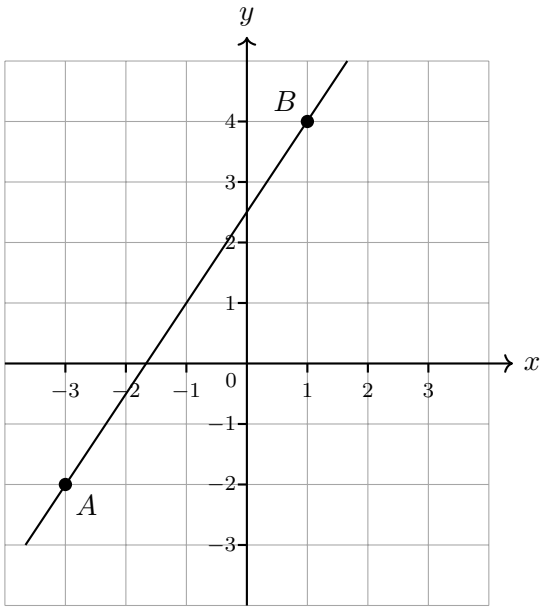
(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(0, c)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of c.

$c =$ ..... **[2]**

5 **(a)** The diagram shows a straight line passing through points A and B.
Work out the gradient of the line AB.

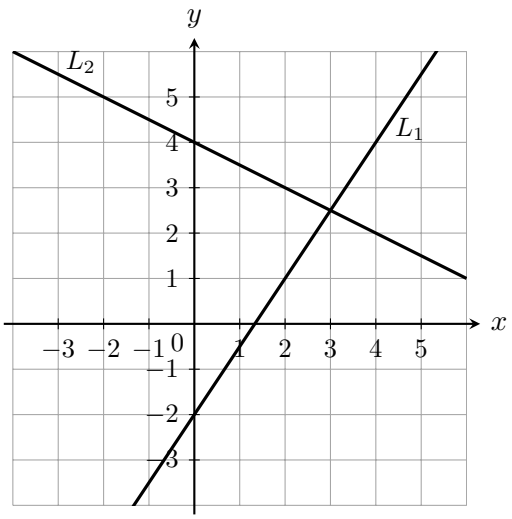


Gradient = ... **[2]**

(b) Point C has coordinates $(-2, 5)$ and point D has coordinates $(4, -7)$.
Calculate the gradient of the line CD.

Gradient = ... **[2]**

6 The diagram shows two straight lines, L_1 and L_2, drawn on a coordinate grid.



(a) Find the gradient of line L_1.

..... **[2]**

(b) Find the gradient of line L_2.

..... **[2]**

(c) A straight line L_3 passes through the points $P(-3, 7)$ and $Q(5, -9)$.
Calculate the gradient of line L_3.

..... **[2]**

7 **(a)** A straight line passes through the points $P(2, k)$ and $Q(5, 3k - 1)$.

(i) Find an expression for the gradient of the line PQ in terms of k.
Give your answer in its simplest form.

Gradient = ... **[2]**

(ii) The gradient of the line PQ is 4.
Find the value of k.

k = ... **[2]**

8 The points $A(-2, 5)$, $B(4, k)$, and $C(10, -1)$ lie in a coordinate plane.

(a) Given that the gradient of the line segment AB is $-\frac{1}{2}$, calculate the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), show that the points A, B, and C lie on the same straight line.

[2]

(c) Point D has coordinates $(p, 3p - 1)$.
Find the value of p for which the line AD has a gradient of 2.

$p =$ ..... **[3]**

9 **(a)** Point P has coordinates $(2a, 3a)$ and point Q has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line PQ in terms of a.
Give your answer in its simplest form.

..... **[2]**

(b) The line PQ is perpendicular to a line with gradient 3.
Work out the value of a.

$a =$ ..... **[3]**

(c) The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of c.

$c =$ ..... **[3]**

10 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a + 6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

11 The points $A(1, 2)$, $B(5, k)$ and $C(k, 2k)$ are three distinct collinear points, where k is a constant.

(a) Write down an expression, in terms of k, for the gradient of the line segment AB.

..... **[1]**

(b) By finding an expression for the gradient of BC, show that $k^2 - 11k + 10 = 0$.

[3]

(c) Solve the equation to find the two possible values of k.

$k =$ ..... , k =

..... **[2]**

(d) Calculate the exact gradient of the line passing through point B and the origin $(0, 0)$.

..... **[2]**

12 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

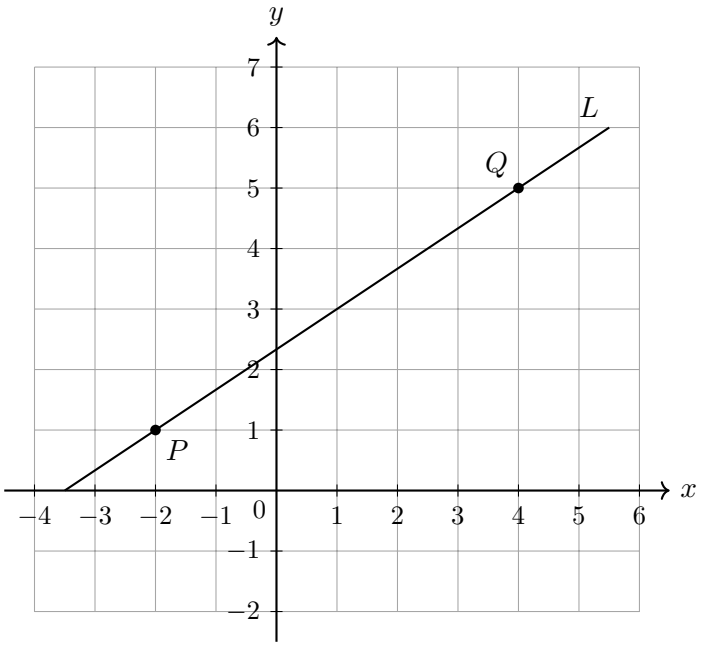
(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

13 **(a)** The diagram shows a straight line L passing through the points $P(-2, 1)$ and $Q(4, 5)$.
Find the gradient of line L.



..... **[2]**

(b) A different line passes through the points $(3, k)$ and $(8, -2)$.
The gradient of this line is -1.4.
Calculate the value of k.

$k =$ ..... **[3]**

14 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

15 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**

16 A straight line passes through the points $P(3k, 4)$ and $Q(k-2, 8)$.
The gradient of this line is $-\frac{1}{2}$.

(a) Find the value of k.

$k =$ ..... **[3]**

(b) Write down the coordinates of P and Q.

$P =$ ..... , $Q =$ ..... **[1]**

(c) Another point R lies on the y-axis such that the gradient of the line passing through P and R is 1.
Find the coordinates of R.

(..... , ...) **[3]**

17 Points $A(-1, -4)$ and $B(3, 8)$ lie on a straight line.

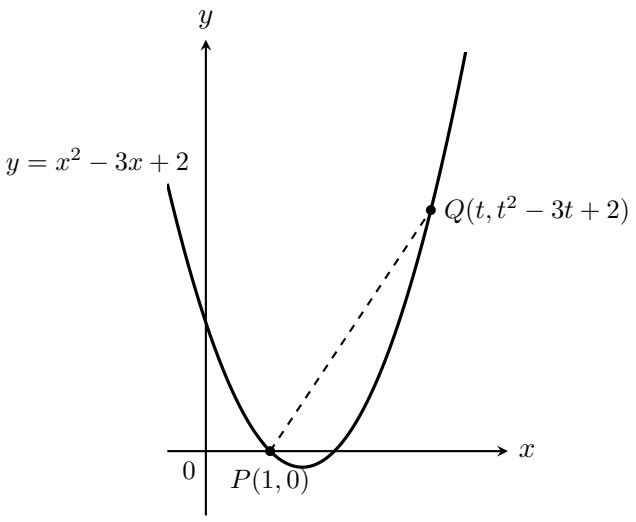
(a) Point $C(p, 23)$ also lies on the line passing through A and B.
Find the value of p.

$p =$ ..... **[3]**

(b) Point D has coordinates $(q, 2q - 1)$.
The gradient of the line AD is equal to $-\frac{1}{3}$.
Find the value of q.

$q =$ ..... **[3]**

18 The diagram shows a sketch of the curve $y = x^2 - 3x + 2$ and two points, $P(1, 0)$ and $Q(t, t^2 - 3t + 2)$, on the curve, where $t > 1$.



(a) Show that the gradient of the chord PQ simplifies to $t - 2$.

[3]

(b) Find the coordinates of point Q when the gradient of the chord PQ is 2.8.

$Q = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

(c) Point R has coordinates $(3.2, -1.5)$.
Find the value of t, where $t > 1$, for which the gradient of QR is equal to the gradient of PQ.
Give your answer correct to 3 significant figures.

$t =$ ..... **[4]**